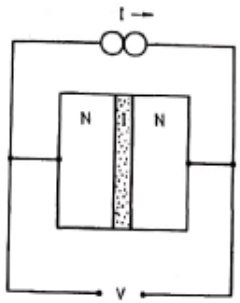
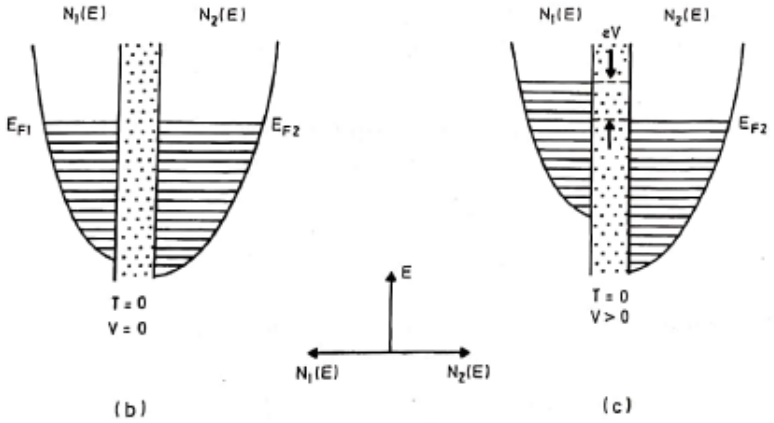
Tunnelling between a normal metal- Insulator normal metal



(a)



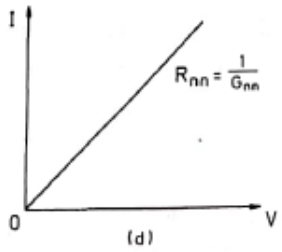


Figure 6.1: The NIN junction: a) Schematic diagram; b) Density of states vs. energy at $T = 0$ in thermal equilibrium; c) The junction with an applied bias voltage: electrons can tunnel from left to the right; d) I-V curve of an NIN junction.

Tunnelling between a normal metal- insulator - superconductor

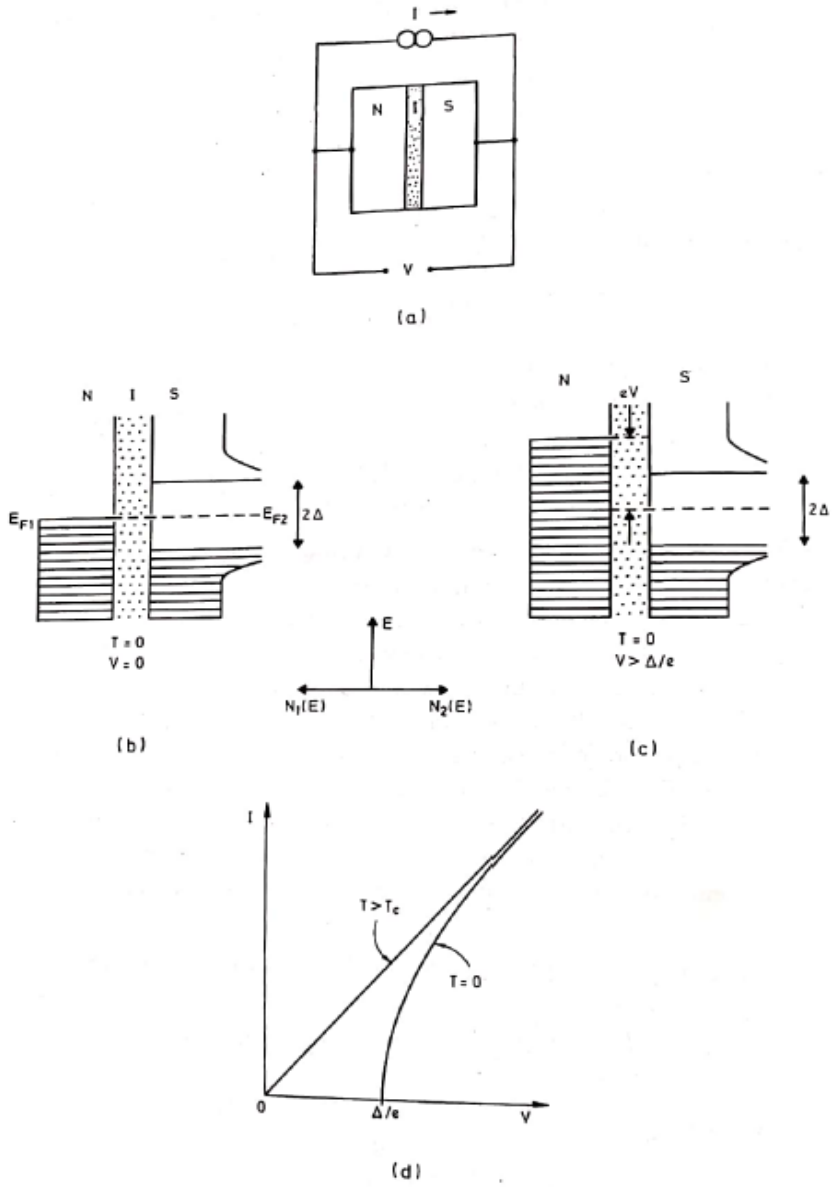


Figure 6.2: The NIS junction: a) Schematic diagram; (b) Density of states vs. energy at $T = 0$ in thermal equilibrium; c) The junction with an applied bias voltage: electrons can tunnel when $V > \frac{\Delta}{e}$; d) I-V characteristics of an NIS junction.

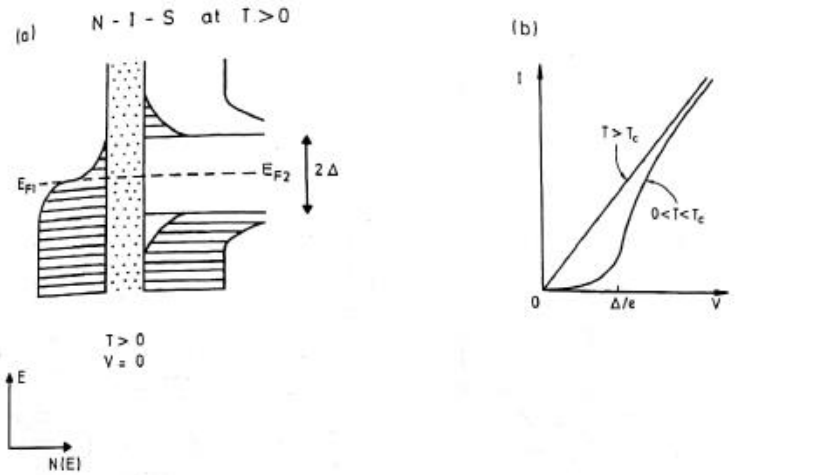
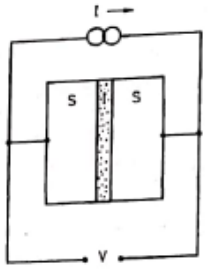
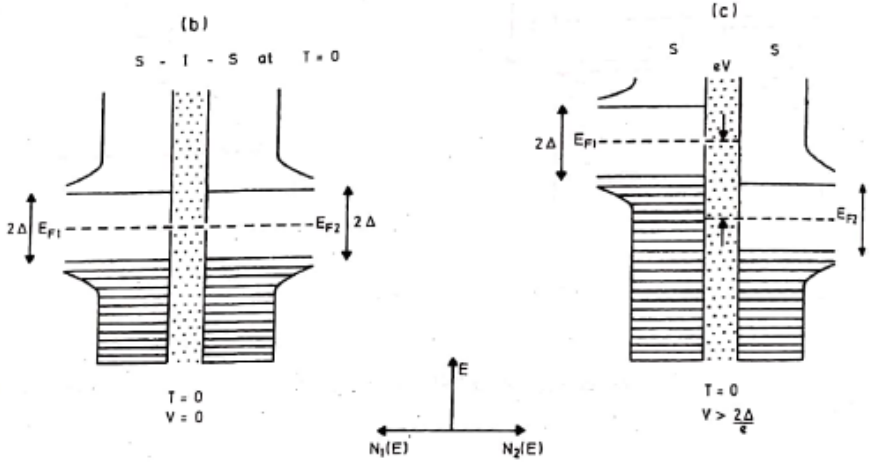
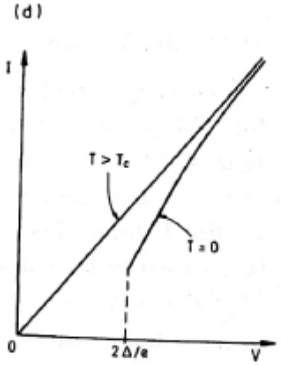


Figure 6.3: Density of states vs. energy of an NIS junction at $T > 0$. b) I-V characteristics for $0 < T < T_c$ and $T > T_c$.

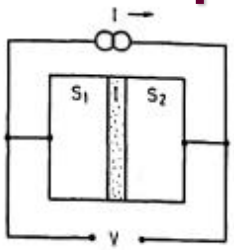
Tunnelling between a superconductor- insulator - superconductor



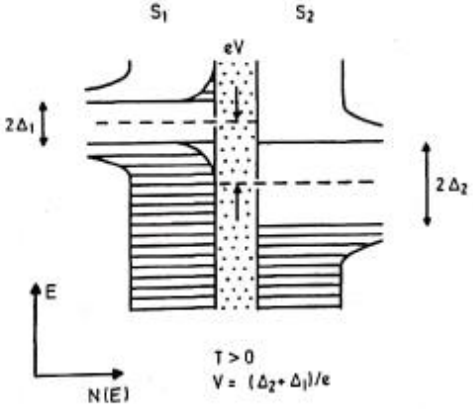




ıre 6.4: The SIS junction: Schematic diagram; b) Density of states vs. energy at $T = 0$ in ther- equilibrium; c) The junction with an applied bias voltage $V > \frac{2\Delta}{e}$; d) the I-V characteristics ı SIS junction for $T = 0$ and $T > T_c$.



(b)



(c)

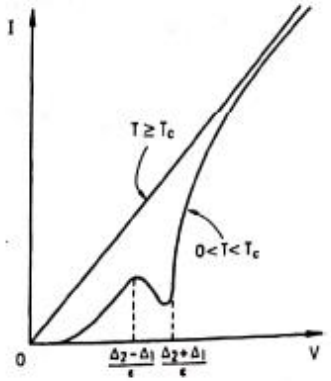
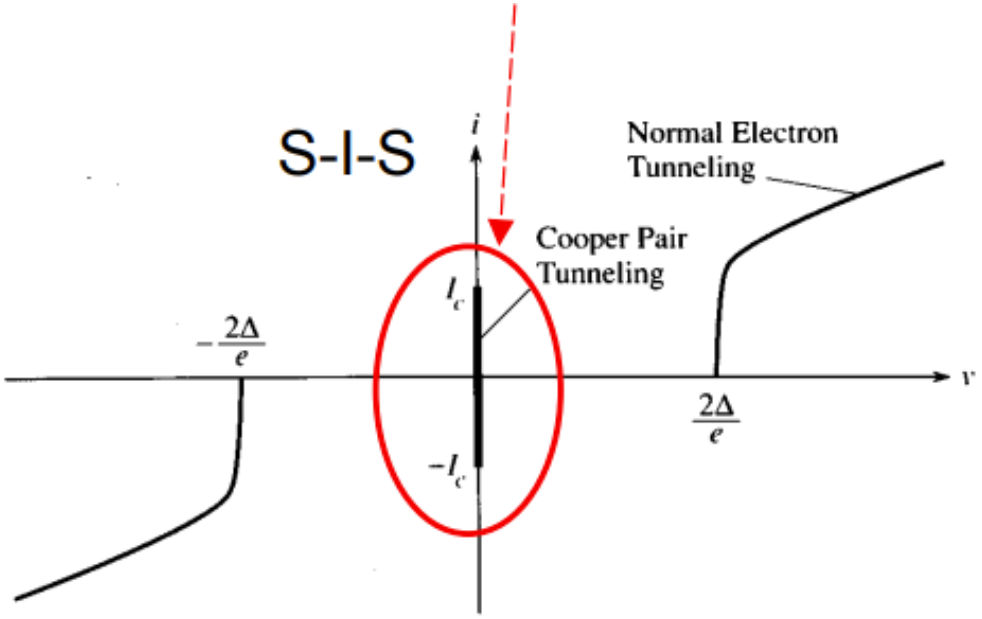
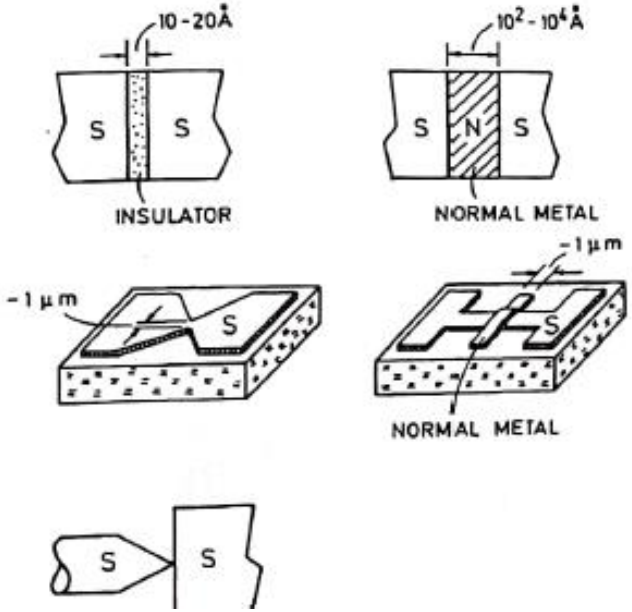


Figure 6.5: The asymmetric SIS junction at finite temperatures: a) Schematic diagram; b) Density of states vs. energy with a bias voltage; c) the corresponding I-V characteristics for $T \gtrsim T_c$ and $0 < T < T_c$.

Tunnelling between a superconductor- insulator - superconductor





Common types of weak link systems are :

i) **SIS junction:** an oxide or other insulator separates two superconducting films. This is the most widely used junction for electronic devices since its preparation can be controlled by thin film fabrication techniques as in semiconductor technology.

ii). **SNS or proximity junction:** thin layer of a normal metal evaporated between two superconducting films.

iii) **Point contact** junction between two bulk superconductors. Typically a superconducting wire is ground to a point, the surface is allowed to get oxidized, and then the point is pressed against a piece of bulk superconductor. Point contacts are set up in such a way that the pressure can be adjusted from outside in order to obtain the required current.

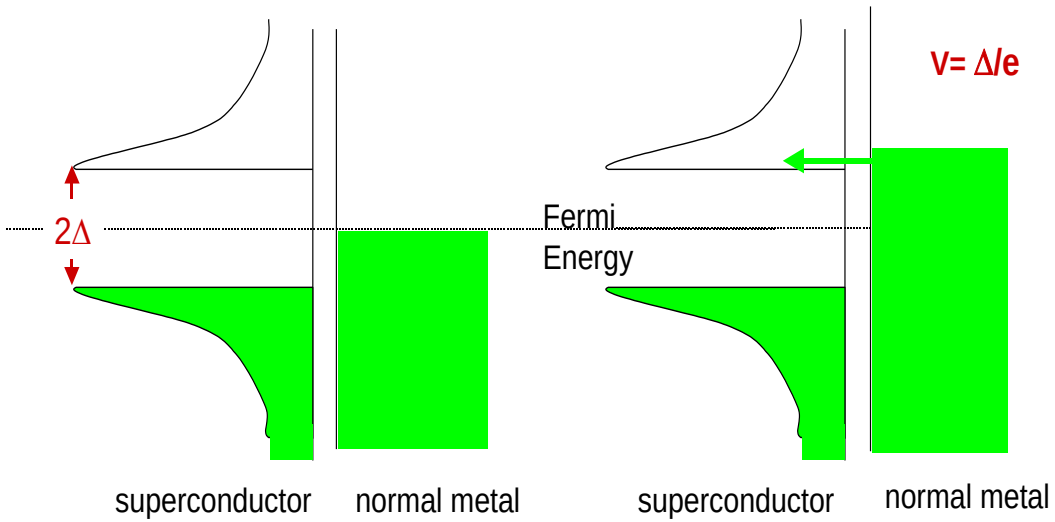
iv) Thin film **microbridge** formed by suitable processing of a superconducting film into a narrow constriction.

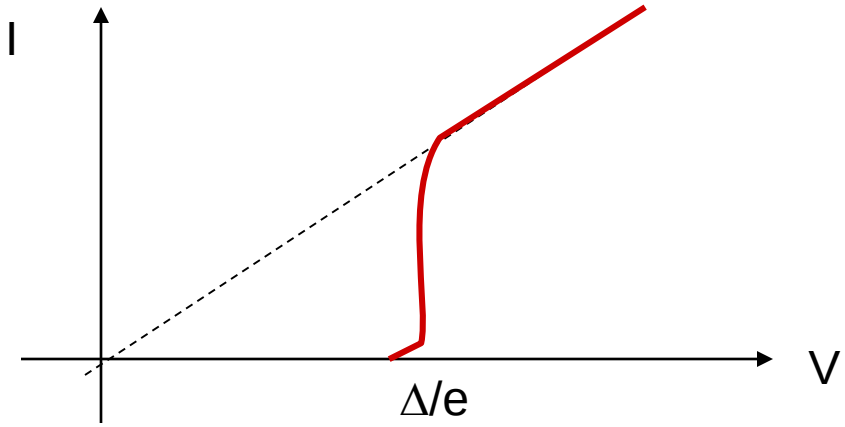
Josephson effects

We have already noted tunneling effects between a superconductor and a normal metal (through an insulating barrier)

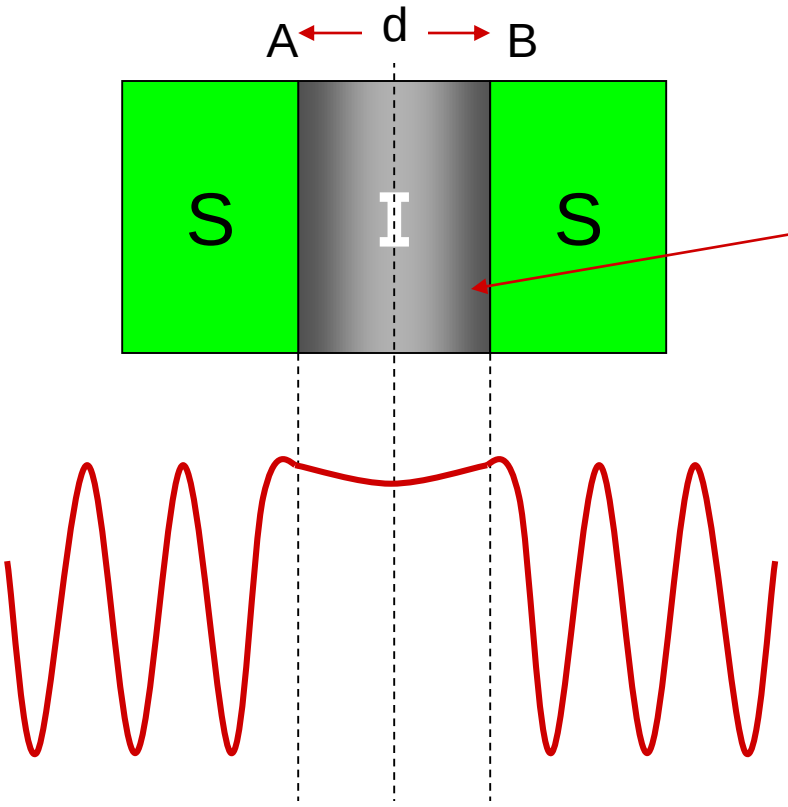
This provides evidence of a superconducting energy gap

Josephson demonstrated tunneling between two superconductors separated by an insulating barrier





The dc Josephson effect



Insulating weak link

Macroscopic wave function

assume that the barrier is sufficiently thin, ie $d \to 0$ and

$$|\Psi|^2 = \frac{n_s}{2} \to 0$$

The dc Josephson effect

The phase difference between any two points along the wavefunction is

$$\Delta\varphi_{12} = \varphi_1 - \varphi_2 = 2\pi\int_1^2 \frac{\hat{x}}{\lambda}\,dl$$

whilst the supercurrent density is

$$J_s = 2e \cdot \frac{n_s}{2} \cdot v$$

So, with $\lambda=h/2mv$, we have $\lambda=hn_se/2mJ_s$, and

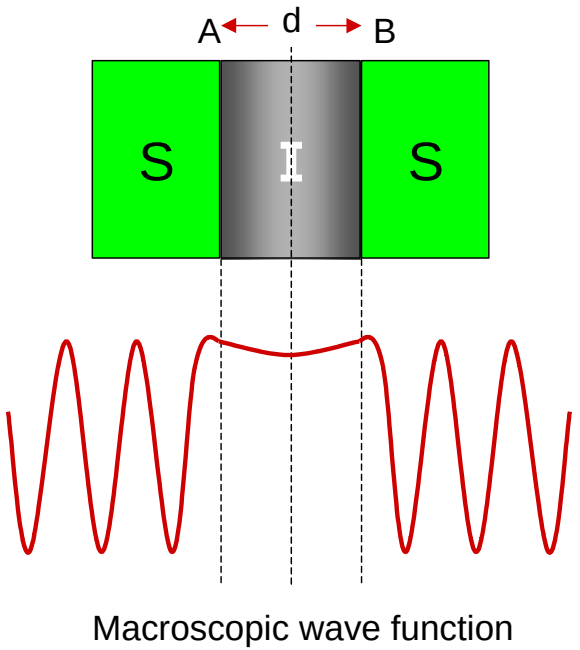
$$\Delta\varphi_{12} = \frac{4\pi m}{hn_s e}\int_1^2 \underline{J}_s\,d\underline{l}$$

ie $$J_s = \frac{\hbar n_s e}{2m}\nabla\varphi$$ or, along the x-direction

$$J_x = \frac{2e\hbar}{2m}|\Psi|^2\frac{d\varphi}{dx}$$

The dc Josephson effect

$$J_x = \frac{2e\hbar}{2m}|\Psi|^2 \frac{d\varphi}{dx}$$



Macroscopic wave function

Assuming that both J_x and $|\psi|^2$ are both constant across the link AB

$$J_x = J_o\gamma$$

where γ is the difference in phase $\phi_B - \phi_A$ and

$$\frac{d\varphi}{dx} = \frac{\varphi_B - \varphi_A}{d}$$

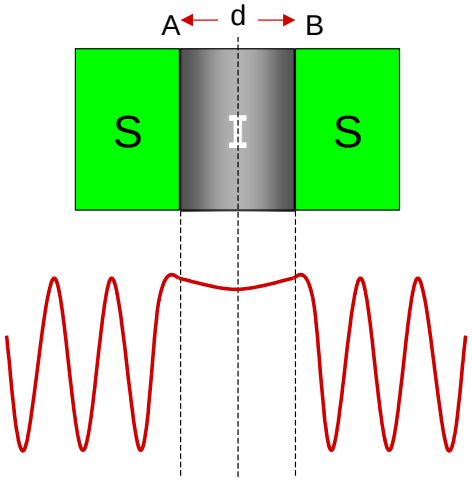
and $$J_o = \frac{2e\hbar}{2md}|\Psi|^2$$

Now as $\phi_B - \phi_A$ and $\phi_B - \phi_A + 2\pi n$ are indistinguishable, we have

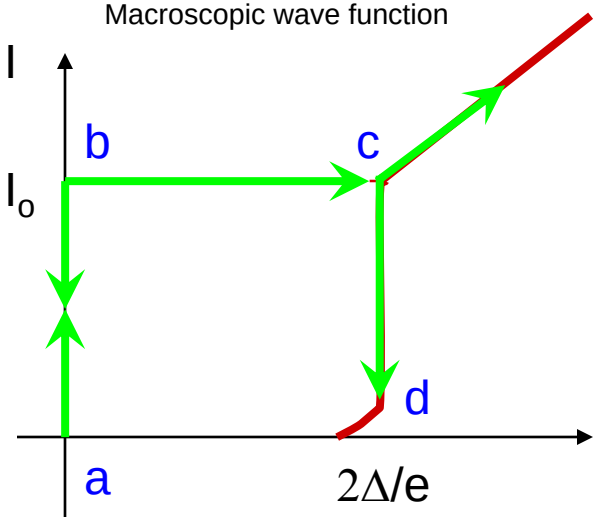
$$J_x = J_o \sin\gamma$$

where J_o is the "*critical current*" of the "superconducting" insulator

The dc Josephson effect



Macroscopic wave function



The insulating weak link may be a bridge of area A for which $J_o=I_o/A$

So below I_o the supercurrent will dominate the transport across the weak link, but above I_o the current is entirely normal

At I_o the device crosses from operation at b to operation at c

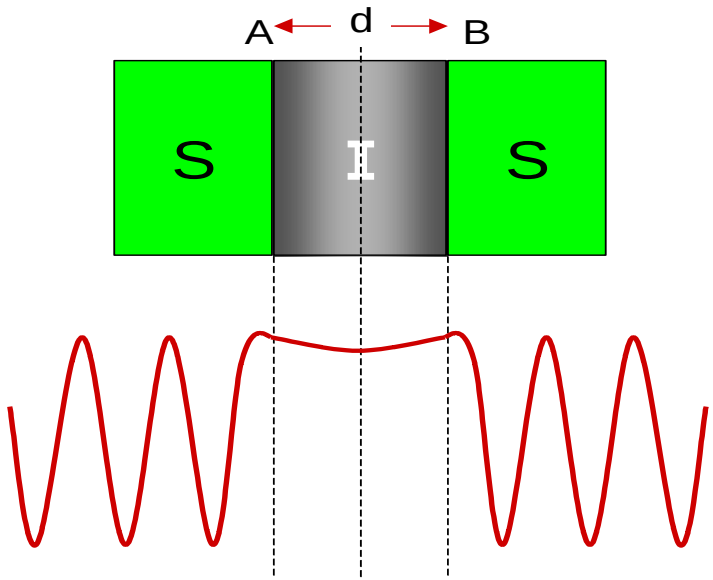
To increase the current above I_o the potential difference across the junction must be increased

If the current is then reduced from point c it follows the curve to point d: *The Josephson junction is hysteretic*

Below I_o there is a supercurrent, but because there is a potential difference across the junction the current oscillates rapidly

- only the dc component is shown here

The dc Josephson effect



Macroscopic wave function

Typically I_0 ~1mA

This dc Josephson device can be used as a switch in logic circuits

It is 10^3 times faster than the semiconducting equivalent, and dissipates 0.01% of the power

The ac Josephson effect

If we now apply a voltage, V, while $I<I_o$ such that the energy difference across the junction is $\Delta E=2eV$, there will be an additional time dependent phase difference across the junction such that:

$$J_x = J_o \sin(\gamma - 2eVt/\hbar)$$

The supercurrent density therefore oscillates at a frequency of $2eV/\hbar$

If the voltage difference is maintained this will correspond to emission of radiation of this frequency

Photon emission is necessary for conservation of energy, because unless is $\Delta E=2eV$ emitted by the electron pairs they will be unable to tunnel into pair states on the other side of the barrier

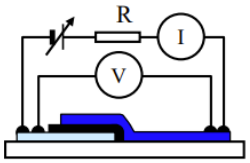
For $V=10^{-6}$ volts, f=484Mhz, $\lambda=620\mu m$ - *independent of the superconducting material*

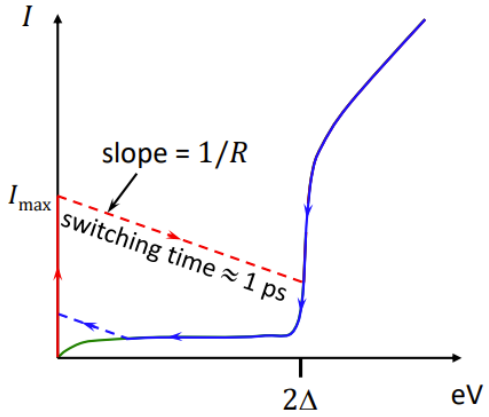
Note: this relates, for the first time, voltage to frequency directly through fundamental constants. It is now used to define the Volt

- ***Josephson effect***: $V = \frac{h}{2e} \cdot f = \Phi_0 \cdot f$ (relation between voltage and time/frequency by flux quantum)
- ***Single electron pump***: $I = e \cdot f$ (relation between current and time by charge quantum)
- ***Quantum Hall effect***: $V = \frac{h}{e^2} \cdot I = R_K \cdot I$ (relation between voltage and current by quantum resistance, unit = 1 Klitzing)

Josephson junction as fast switching device

- $V = 0$: Josephson current
- $V \neq 0$: quasiparticle current



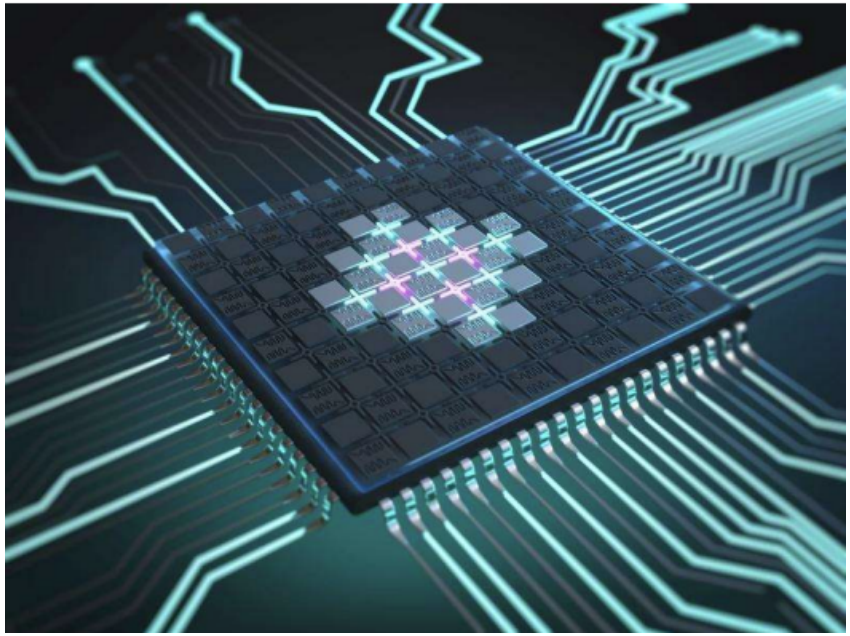


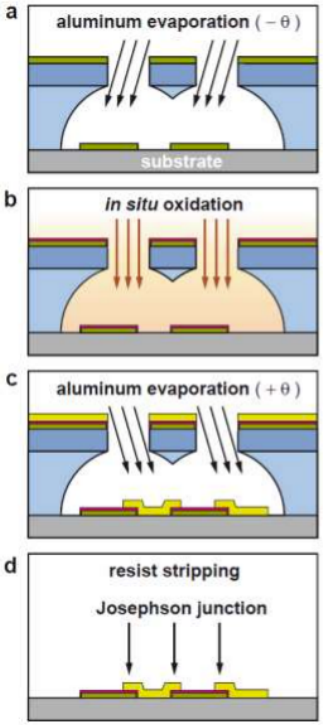
- hysteresis:

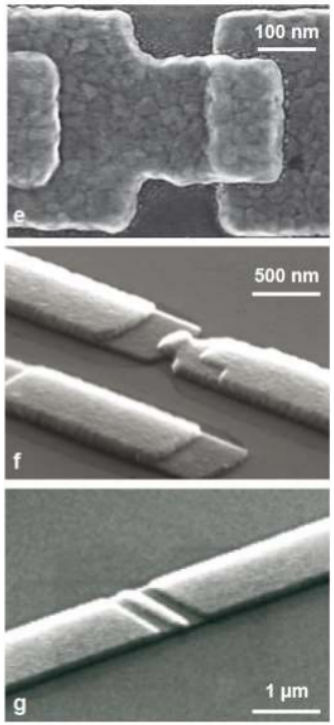
  fast switching device

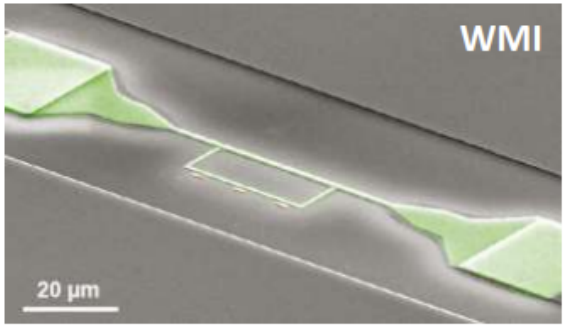
  very low power consumption

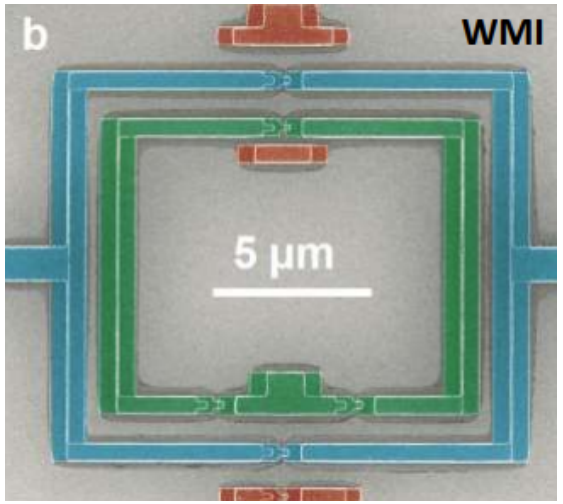
  ⇒ **Josephson digital electronics**











The Superconducting QUantum Interference Device

DC SQUID: superconducting ring containing two Josephson junctions (of critical current I_c) in parallel

Assume flux threading each separate junction is negligible

The total current flowing into the circuit is

$$I = I_c(\sin\gamma_A + \sin\gamma_B)$$

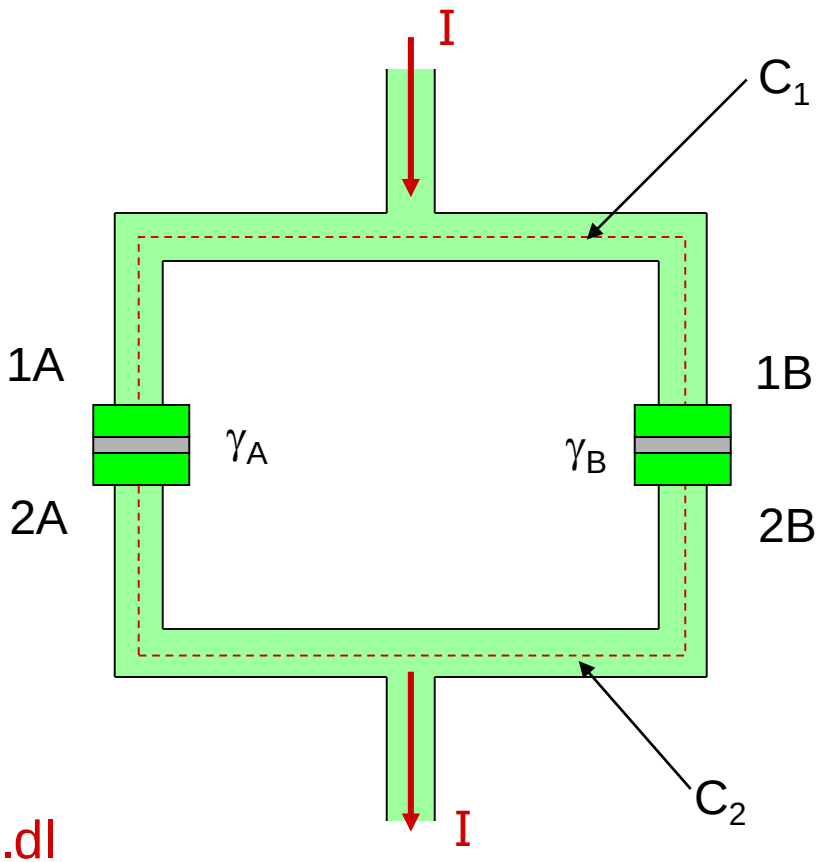
where

$$\gamma_A = \phi_{2A} - \phi_{1A} \qquad \gamma_B = \phi_{2B} - \phi_{1B}$$

Remember that the total phase difference is:

$$(\Delta\varphi)_{XY} = \frac{2\pi m^*}{h n_s^* e^*}\int_X^Y \underline{J_S}.\underline{dl} + \frac{2\pi e^*}{h}\int_X^Y \underline{A}.\underline{dl}$$

Zero in centre of superconductor ← *Phase difference due to current* — *Phase difference due to change of flux density*



The Superconducting QUantum Interference Device

So the phase difference between two points *inside* the superconductor is:

$$(\Delta\varphi)_{XY} = \frac{2\pi e^*}{h}\int_X^Y \underline{A}.\underline{dl}$$

and the phase differences along C_1 and C_2 are

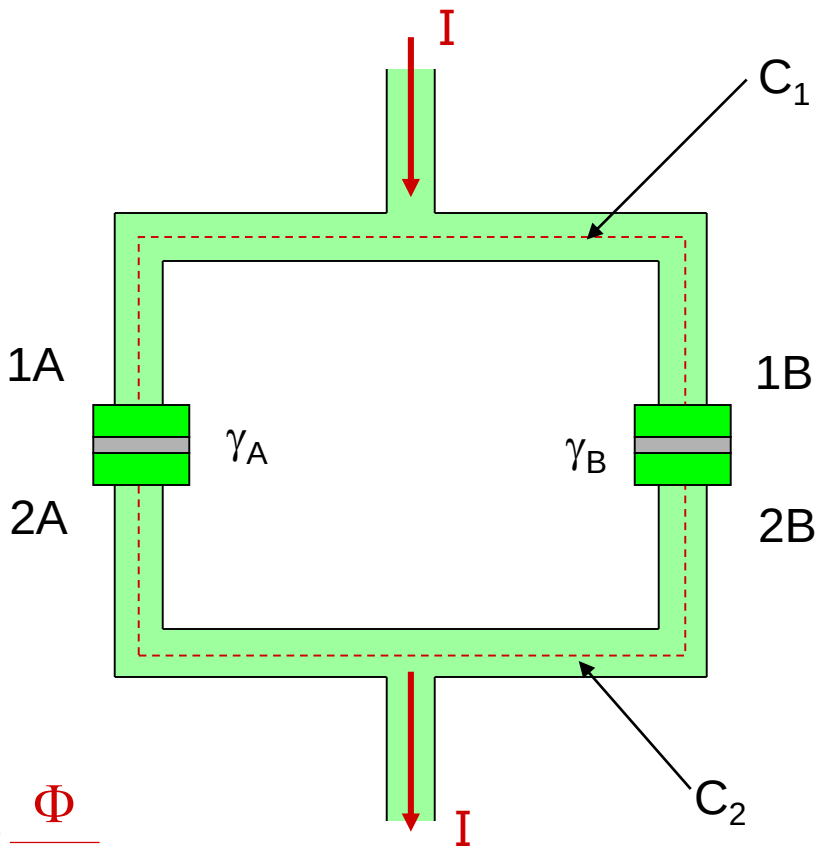
$$\varphi_{1B} - \phi_{1A} = \frac{2\pi e^*}{h}\int_{1A}^{1B} \underline{A}.\underline{dl}$$

$$\varphi_{2A} - \phi_{2B} = \frac{2\pi e^*}{h}\int_{2B}^{2A} \underline{A}.\underline{dl}$$

Summing these equations

$$\varphi_{1B} - \phi_{1A} + \varphi_{2A} - \phi_{2B} = \frac{2\pi e^*}{h}\oint \underline{A}.\underline{dl} \quad = 2\pi\frac{\Phi}{\Phi_o}$$

hence $\gamma_A - \gamma_B = 2\pi\dfrac{\Phi}{\Phi_o}$ where Φ is the flux threading the SQUID loop



The Superconducting QUantum Interference Device

if $\gamma_A - \gamma_B = 2\pi \frac{\Phi}{\Phi_o}$ we can write

$$\gamma_A = \gamma_o + \pi \frac{\Phi}{\Phi_o} \qquad \gamma_B = \gamma_o - \pi \frac{\Phi}{\Phi_o}$$

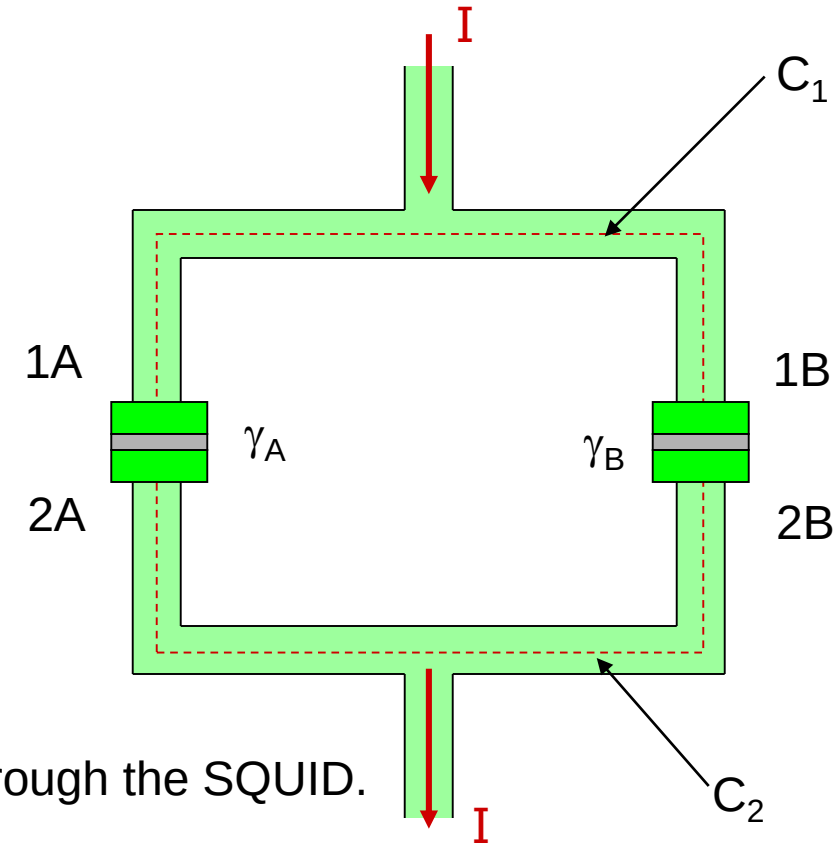
So $I = I_c(\sin\gamma_A + \sin\gamma_B)$ becomes

$$I = 2I_c \sin\gamma_o \cos\pi \frac{\Phi}{\Phi_o}$$

$$= I_{max} \sin\gamma_o$$

where

$$I_{max} = 2I_c \cos\pi \frac{\Phi}{\Phi_o}$$



I_{max} is the maximum current that can flow through the SQUID.

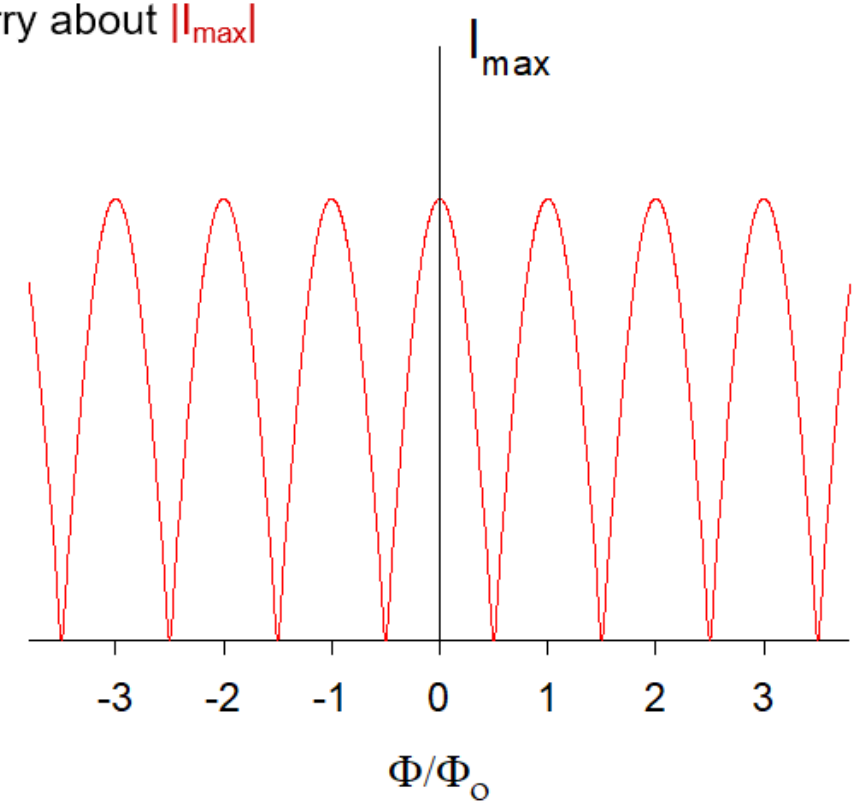
This "critical current" is modulated by the magnetic flux trapped between the junctions with a period defined by the flux quantum

As long as the current is smaller than Imax, the phase y0 adjusted itself so the current flow without dissipation.

The DC SQUID

As we are only concerned with the current necessary to drive the SQUID normal

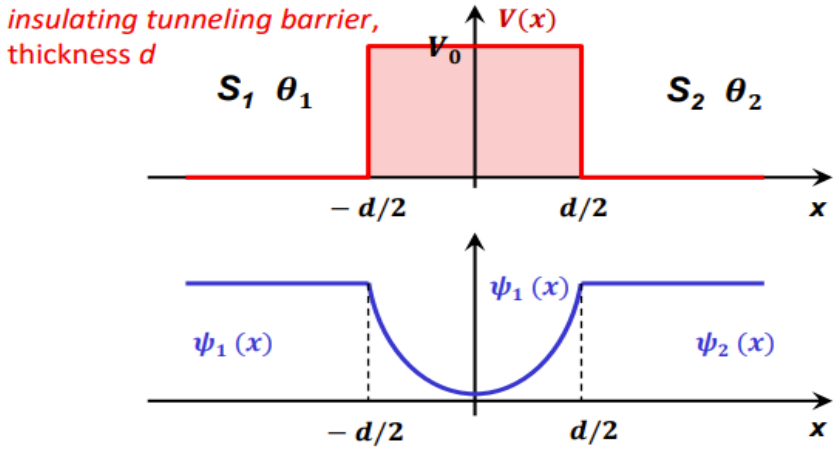
As we are only concerned with the current necessary to drive the SQUID normal
we need only worry about $|I_{max}|$



If the area of loop is 1cm^2
The flux periodicity corresponds to magnetic field

SQUIDs are therefore capable of measuring *extremely* small changes in magnetic flux, much less than that associated with a single flux quantum

Josephson effect in superconducting tunnel junctions



- **derive the Josephson equations**
  - starting point is time-dependent Schrödinger equation

$$\imath\hbar\frac{\partial\psi(\mathbf{r},t)}{\partial t} = E\,\psi(\mathbf{r},t)$$

  - S_1 and S_2 are described by macroscopic wave functions

$$\psi_1(\mathbf{r},t) = \psi_{01}\,e^{\imath\theta_1(\mathbf{r},t)} \qquad n_{s1} = |\psi_{01}|^2$$

$$\psi_2(\mathbf{r},t) = \psi_{02}\,e^{\imath\theta_2(\mathbf{r},t)} \qquad n_{s2} = |\psi_{02}|^2$$

  - finite coupling between S_1 and S_2 is introduced by small perturbation T (tunnel coupling)

$$\imath\hbar\frac{\partial\psi_1(\mathbf{r},t)}{\partial t} = E_1\,\psi_1(\mathbf{r},t) + T_{\mathrm{LR}}\psi_2(\mathbf{r},t) = +e\Delta\phi\;\psi_1(\mathbf{r},t) + T_{\mathrm{LR}}\psi_2(\mathbf{r},t)$$

$$\imath\hbar\frac{\partial\psi_2(\mathbf{r},t)}{\partial t} = E_2\,\psi_2(\mathbf{r},t) + T_{\mathrm{RL}}\psi_1(\mathbf{r},t) = -e\Delta\phi\;\psi_2(\mathbf{r},t) + T_{\mathrm{RL}}\psi_1(\mathbf{r},t)$$

$$\Delta\phi = \frac{E_1 - E_2}{|q_s|} = \frac{E_1 - E_2}{2e} = \frac{1}{2}\frac{E_1 - E_2}{e}$$

– by inserting the wave functions ψ_1, ψ_2 into the time-dependent Schhrödinger equation we obtain for the **imaginary part**:

$$\frac{\partial\theta_1(t)}{\partial t} = -\frac{T}{\hbar}\sqrt{\frac{n_{s2}}{n_{s1}}}\cos\varphi(t) - \frac{e\Delta\phi}{\hbar}$$

$$\frac{\partial\theta_2(t)}{\partial t} = -\frac{T}{\hbar}\sqrt{\frac{n_{s1}}{n_{s2}}}\cos\varphi(t) + \frac{e\Delta\phi}{\hbar}$$

we use $T_{\mathrm{LR}} = T_{\mathrm{RL}} = T$ and $\varphi = \theta_2 - \theta_1$,

– for the **real part** we obtain:

$$\frac{\partial n_{s1}(t)}{\partial t} = +\frac{2T}{\hbar}\sqrt{n_{s1}n_{s2}}\sin\varphi(t)$$

$$\frac{\partial n_{s2}(t)}{\partial t} = -\frac{2T}{\hbar}\sqrt{n_{s1}n_{s2}}\sin\varphi(t)$$

we see that $\frac{\partial n_{s1}(\mathbf{r},t)}{\partial t} = -\frac{\partial n_{s2}(\mathbf{r},t)}{\partial t} \Rightarrow$ ***conservation of particle number***

– supercurrent density:

$$J_s^{1\to2} = \frac{2e}{A}\frac{\partial n_{s1}(t)}{\partial t}$$

$$J_s^{2\to1} = \frac{2e}{A}\frac{\partial n_{s2}(t)}{\partial t}$$

$$J_s = J_s^{1\to2} - J_s^{2\to1} = \frac{4eT}{\hbar A}\frac{\partial n_{s2}(t)}{\partial t}\sqrt{n_{s1}n_{s2}}\sin\varphi(t) = J_c\sin\varphi(t)$$

1st Josephson equation